Wide Projects (Application of big data technologies)

Wide projects require you to use **at least 3 different big data technologies** to perform an interesting data analytical task and/or build a service. The goal is to demonstrate "Breadth" in architecting a data pipeline.

Description:

- Combine technologies to solve a real-world problem (e.g., Real-time recommendation, Log analysis dashboard).
- Deploying your app on the cloud (AWS/Azure/GCP) is highly encouraged.

Technological Stack (Examples):

Each of the following counts as one "technology":

- **Compute:** Spark RDD, Spark SQL, Spark Streaming.
- **Storage/NoSQL:** MongoDB, HBase, Cassandra, Redis.
- **Messaging:** Kafka, RabbitMQ.
- **Graph:** GraphX, GraphFrames, Neo4j.
- **ML:** MLLib, PyTorch (Distributed).
- **Visualization:** Kibana, Grafana, Tableau (connected to backend).

Sample Data Sources:

- IMDB Dataset of 50K Movie Reviews
- Climate Change: Earth Surface Temperature Data
- NYC Taxi Trip Data
- Many data sets are publicly available (such as on Kaggle and UCI), and can be used in combination to perform some interesting analytics.

Requirements:

- The proposal of a wide project should contain the following contents:
  1. Description of the task to be performed.
  2. Description of the planned datasets.
  3. Planned technologies to use.
- The presentation of a wide project should contain the following contents:
  1. Description of the task.
  2. Description of the used datasets.
  3. System architecture (showing all technologies and their relationships).
  4. Obtained results (or a demo if the project provides a service).
  5. Discussion on the benefits of big data technologies over traditional methods.
- The final report should contain all the above contents in the presentation, as well as the following additional materials:
  1. The source code of your implementation (as a separate file) .
  2. References.
- Wide projects will be graded based on the following criteria (by the order of importance):
  1. System complexity and synergy of different technologies.
  2. Integration of different data types (structured/semi-

structured/unstructured, batch vs streaming).
3. User interface and system demonstration.
4. Performance and accuracy.

References:

- Sample wide project reports from previous semesters: 1, 2.